MIL-C-27802A(USAF)
9 September 1965
Superseding
MIL-C-27802(USAF)
28 May 1965

MILITARY SPECIFICATION

CONTROL-MONITOR OA-4106( )/MCC-12

1. SCOPE

1.1 This specification covers one type of control-monitor, designated Control-Monitor OA-4106( )/MCC-12 hereinafter called "group pilot alarm".

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-4158	Electronic Equipment Ground; General Requirements For
MIL-N-7513	Nomenclature Assignment And Nameplate Approval; Contractors Method For Obtaining
MIL-F-14072	Finishes For Ground Signal Equipment
MIL-R-26474	Reliability Requirements For Production Ground Electronic Equipment
MIL-M-26512	Maintainability Requirements For Aerospace Systems and Equipment
MIL-R-27542	Reliability Program Requirements For Aerospace Systems, Subsystems and Equipment
MIL-M-27741	Multiplexer Set AN/FCC-17( ), AN/FCC-21( ), AN/FCC-22( ) and AN/MCC-12( ) (Mobile And Fixed Installations)
MIL-D-27801	Demultiplexer Group OA-4104( )/GCC
MIL-P-27810	Power Supply Set, OA-4109( )/MCC-12 and Power Supply Set, OA-6830( )/FCC-17
MIL-I-27820	Indicator-Power Supply ID-1077( )/MRC-98 and Control Indicator C-3668( )/MCC-12
MIL-H-27894	Human Engineering Requirements For Aerospace Systems and Equipment

STANDARDS

Federal

FED-STD-595	Colors

FSC 5805

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Military

MIL-STD-449	Radio Frequency Spectrum Characteristics, Measurements Of
MIL-STD-778	Definitions Of Terms For Maintainability
MIL-STD-826	Electromagnetic Interference Test Requirements And Test Methods
MIL-STD-831	Test Reports, Preparation Of

DRAWINGS

Air Force

DL-790-01351	Control-Monitor OA-4106( )/MCC-12

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

The Clearing House for Federal Scientific and Technical Services (CFSTS)

PB-161894	RADC Reliability Notebook RADC-TR-58-111

(Copies of CFSTS documents may be obtained from the Clearing House for Federal Scientific and Technical Services (CFSTS) U.S. Dept. of Commerce, Washington 25, D.C. or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 Components.- The group pilot alarm shall consist of:

Item No.	Quantity	Description	Requirement
1	1-10 each as req'd	Monitor, RF	3.6.1

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3.3 General specification.- The requirements of MIL-E-4158 and MIL-M-27741 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3.1 Service conditions.- Service conditions shall conform to MIL-E-4158 and MIL-M-27741 except as specified below.

3.3.1.2 Electrical.- The group pilot alarm shall be designed to operate from a minus 48 volt, direct current (VDC) power source. (See MIL-P-27810)

3.3.2 Interference.- The group pilot alarm shall conform to the limits specified for Class G_D equipment in accordance with MIL-STD-826.

3.3.3 Structural welding.- Structural welding shall be in accordance with MIL-E-4158. The requirement for qualification of welders shall not apply.

3.3.4 Human engineering.- In addition to the requirements of MIL-E-4158 the contractor shall apply human engineering design principles which shall maximize the military effectiveness of man-equipment combinations, and minimize unnecessary demands upon human skill, training, and manpower. This requirement shall be accomplished by the utilization of the human engineering requirements specified in MIL-H-27894.

3.3.5 Measurements of radio frequency spectrum.- Measurements of radio frequency spectrum characteristics and plans for collection shall be provided in accordance with MIL-STD-449.

3.3.6 Nomenclature and nameplate approval.- When required in the contract nomenclature and nameplate approval shall be obtained in accordance with the requirements of MIL-N-7513. (See 6.2.g) The parentheses appearing in the nomenclature may be dropped, or replaced by a letter, to designate the particular design.

3.3.7 Maintainability program.- The required maintainability shall be achieved through a maintainability program performed in accordance with MIL-M-26512 and Appendix A thereto. The terms and definitions for maintainability not otherwise described or delineated herein shall be in accordance with MIL-STD-778.

3.3.7.1 Maintainability.- The group pilot alarm shall possess a mean time to repair (MTTR) of no greater than 15 minutes.

3.3.8 Reliability program.- The required reliability shall be obtained through a reliability program performed in accordance with MIL-R-27542 and MIL-R-26474 with exceptions or additions as follows:

a. PB-161894 and revisions to date of award shall be utilized in lieu of AD11427 and AD148556 where the latter are referred to in MIL-R-26474.

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b. The contractor shall perform an analysis of the proposed design for the group pilot alarm to determine compatibility with the required mean time between failure (MTBF). A failure rate shall be assigned to each part in the group pilot alarm in accordance with the data presented in section 8 of the latest revision to RADC Reliability Notebook PB-161894. Parts not included in the coverage of PB-161894 shall be assumed to possess the failure rate of the most similar part in the coverage. Where this in unrealistic, any valid existing data shall be used upon the approval of the procuring activity.

c. When required by the contract a reliability analysis shall be submitted to the procuring activity. The analysis shall include a comparison of the predicted MTBF for the equipment with the required MTBF. (See 6.2.h). Where the predicted figure is less than the requirement, the contractor shall accomplish such changes in design as are necessary to raise the predicted MTBF to the required value.

(1) The reliability analysis shall be revised periodically to consider changes or better data on part stress levels. A revised report shall be issued no later than every 90 days from the submission of the original report. (See 6.2.i)

(2) As the fabrication of each component of the group pilot alarm is completed such that all enclosures are in place and normal equipment performance is possible, the contractor shall prepare and when required by the contract submit to the procuring activity a final prediction of the reliability of the group pilot alarm. (See 6.2.j) If this prediction indicates a lesser reliability than that required, the contractor shall accomplish such changes in design,part application and part stress as are necessary to reduce the final indicated group pilot alarm failure rate to an acceptable level.

(3) When required by the contract a final reliability and maintainability report shall be submitted 30 days after final acceptance of the group pilot alarm by the Air Force and shall include but not be limited to the following: (See 6.2.k)

(a) The predicted reliability and mean down time of the group pilot alarm under operational conditions.

(b) Measurement of reliability/maintainability achieved.

(c) Features of group pilot alarm which are weakest from the standpoint of reliability/maintainability.

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(d) Gains achieved by reliability/maintainability program.

(e) Conditions effecting reliability/maintainability.

3.3.8.1 Reliability.- The group pilot alarm shall possess a MTBF of no less than 1900 hours.

3.4 Design.- The group pilot alarm shall be designed in accordance with drawing DL-790-01351 and shall monitor the 64-KC pilot frequency contained in the 60-108 KC group frequency band input to a number, not to exceed 10, of Demultiplexer Group OA-4104( )/GCC (MIL-D-27801). The group pilot alarm shall then amplify and rectify the 64-KC signal voltage and shall have the following characteristics: (See 6.2.d)

a.	Input frequency	64-KC group pilot
b.	Input impedance	150 ohms
c.	Input level	-50.5 DBM
d.	Source impedance	7.5 ohms
e.	Number of inputs	1 to 10 group pilots
f.	Alarm outputs	10 group pilot alarm lamps (red) 1 major alarm (DC signal)
g.	Alarm cutoff	Switch for major alarm cutoff with yellow indicator lamp
h.	Lamp supply	30-50 VDC or 37 VAC, 60 cycle at 40 MA per lamp
i.	DC supply	48 V at 18 (MA) per group

3.4.1 Input.- The group pilot alarm shall have the following inputs:

	Input Source	Signal	Function
1.	Demultiplexer Group OA-4104( )/GCC	64 KC pilot	Monitor and alarm actuation
2.	Demultiplexer Group OA-4104( )/GCC	48 Volt DC	Power supply
3.	Control-Indicator C-3668( )/MCC-12 (MIL-T-27820) Indicator-Power Supply ID-1077( )/MRC-98 (MIL-T-27820)	40 volts AC 40 volts DC	Alarm lamp power

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3.4.2 Output.- The output of the Indicator-Power Supply ID-1077( )/MRC-98 or Control-Indicator C-3668( )/MCC-12 shall actuate its major alarm lamp.

3.4.3 Front panel controls.- The group pilot alarm shall be provided with 10 alarm cutoff switches, one for each input group of 12 channels, to cutoff the output actuating the master alarm major alarm lamp when operated to the cutoff position.

3.4.4 Relays.- Relay contacts shall complete the circuit to the group pilot alarm lamp indicators and to Indicator Power Supply ID-1077( )/MRC-98 or Control Indicator C-3668( )/MCC-12 when a 64-KC pilot is absent.

3.4.5 Lamp indicators.- Lamp indicators shall be placed on the front of the group pilot alarm and shall indicate the loss of the pilot signal.

3.4.6 Rack mounting.- Slide rails shall be affixed to the group pilot alarm for mounting in a rack.

3.5 Performance

3.5.1 Alarm performance.- Upon failure of one or more of the 64 KC pilot signals from the Demultiplexer Group OA-4104( )/GCC components monitored, the following shall occur:

a. A relay shall actuate to close the circuit to and light the appropriate group pilot alarm lamp.

b. A circuit shall be closed to the Indicator-Power Supply ID-1077( )/MRC-98 panel to actuate its major alarm circuitry.

3.5.2 Alarm cutoff performance.- Operation of the appropriate alarm cutoff switch to the cutoff position shall perform the following:

a. Open the circuit to the Indicator-Power Supply ID-1077( )/MRC-98 disconnect its major alarm circuitry.

b. Close the circuit to the alarm cutoff lamp, causing it to light.

3.6 Details of components.

3.6.1 Radio frequency (RF) monitor.- The RF monitor shall track the presence of the 64-KC pilot frequency, and shall operate an alarm when the pilot fails.

3.6.1.1 The RF monitor shall have the following characteristics:

a.	Resonant frequency	64 K,C.
b.	Output current	
	Input level -50.5 DBM	2.85 MA
	No input signal	0 MA
c.	Power requirements	18 MA at 48 volts DC

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3.6.2 Racks (See MIL-M-27741 para 3.6.1)

3.7 Dimensions.- The overall dimensions of the group pilot alarm shall not exceed the following:

Height	3-½ inches
Width	22-1/8 inches
Depth	20-3/16 inches

3.8 Weight.- The total weight of the group pilot alarm shall not exceed 22 pounds.

3.9 Finish and color.- Unless otherwise specified, (see 6.2.e) finishes shall be in accordance with MIL-F-14072. The color of type I surfaces (as defined in MIL-F-14072) shall be color No. 26239 in accordance with FED-STD-595.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. (See 6.2.c)

4.2 Classification of tests.- The inspection and testing of the group pilot alarm shall be classified as follows:

a. Preproduction testing.....See 4.4
b. Acceptance testing.....See 4.5

4.3 Test conditions.- Unless otherwise specified (see 4.6.4), all tests required by this specification shall be made at prevailing ambient temperature and humidity conditions.

4.3.1 The group pilot alarm shall be tested with the following power:

a. Minus 48 volts DC; 18 MA per group; ripple component less than 20 millivolts (MV).
b. 30 to 50 volts DC or 37 volts AC; 40 MA per group.

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4.4 Preproduction testing

4.4.1 Preproduction test samples.- The preproduction test samples shall consist of models representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. (See 6.2.1)

4.4.2 Preproduction test reports.- When the contractor conducts the preproduction tests, he shall, after completion of the tests, prepare a preproduction test report when required by the contract according to MIL-STD-831, and furnish three complete copies of the report to the procuring activity. (See 6.2.f)

4.4.3 Preproduction tests.- The preproduction tests shall consist of all tests described under 4.6 "Test methods".

4.5 Acceptance tests.- Each group pilot alarm shall be subjected to the acceptance tests. The acceptance tests shall consist of the following tests as described under 4.6 "Test methods" of this specification:

a. Mechanical inspection tests.....(See 4.6.1)
b. Electrical tests.....(See 4.6.2)
c. Performance tests.....(See 4.6.3)

4.6 Test methods

4.6.1 Mechanical inspection tests.- The group pilot alarm shall be given a thorough mechanical and visual inspection and test to determine that the quality of all materials and workmanship is in compliance with the requirements of this specification. Particular attention shall be given to the following:

a. Completeness
b. Nameplates, identification markings and labels
c. Ease of operation of gears, adjustable and sliding parts, thumb screws, controls and switches.
d. Finishes
e. Welded joints
f. Check of solder joints
g. The fit of components in their respective positions
h. Check of mounting means
i. Check of lubrication and rust prevention
j. Check of safety features and interlocks
k. Loose fastening and securing devices or parts
l. Accessibility of components and parts for servicing
m. Cable runs between components including plugs and receptacles

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n. Grounding connections
o. Over-all dimensions check
p. Weight check
q. Other visual defects

4.6.2 Electrical tests.- The group pilot alarm shall be given all electrical tests necessary to confirm that all circuits are inherently sound and in compliance with the requirements of this specification. The electrical tests shall include, but not necessarily be restricted to, the following:

4.6.2.1 Continuity.- Each electrical component and each cord and cable shall be given a continuity test to ascertain if it is wired and connected correctly and that good electrical contact is obtained.

4.6.2.2 Operating voltages.- The operating voltage at all important points shall be checked for conformance to those shown on the circuit label and schematic drawing. This shall be done with all controls set for normal operation of the equipment.

4.6.2.3 Power consumption test.- A power consumption test shall be conducted and shall consist of the following:

a. Power consumption (volts, amperes and power factor.)
b. Total heat dissipation (in British Thermal Units per hour).

4.6.3 Performance test.- The group pilot alarm shall be given all performance tests necessary to confirm that the equipment meets the performance requirements specified herein. The performance tests shall include, but not necessarily be limited to, tests of the following:

a. Alarm output (see 4.6.3.3)
b. Alarm cutoff (see 4.6.3.4)

4.6.3.1 Test equipment.- The following test equipment or equivalent is necessary to test the group pilot alarm.

a. 1 - Oscillator, Hewlett-Packard 200CD
b. 1 - Electronic Counter, Hewlett-Packard 523B
c. 1 - Power Supply, Hewlett-Packard 711A
d. 1 - VTVM, Hewlett-Packard 400H
e. 1 - Volt-Ohm-Milliammeter, Triplett 630NA

4.6.3.2 Connect the equipment as shown in figure 1.

4.6.3.3 Alarm performance test.- Perform tests 1, 2 and 3 indicated on Table 1, and record the results.

4.6.3.4 Alarm cutoff performance.- Perform tests number 4 and 5 indicated on Table 1, and record the results.

4.6.4 Environmental tests.- The group carrier supply shall be subjected to the environmental conditions specified in MIL-M-27741.

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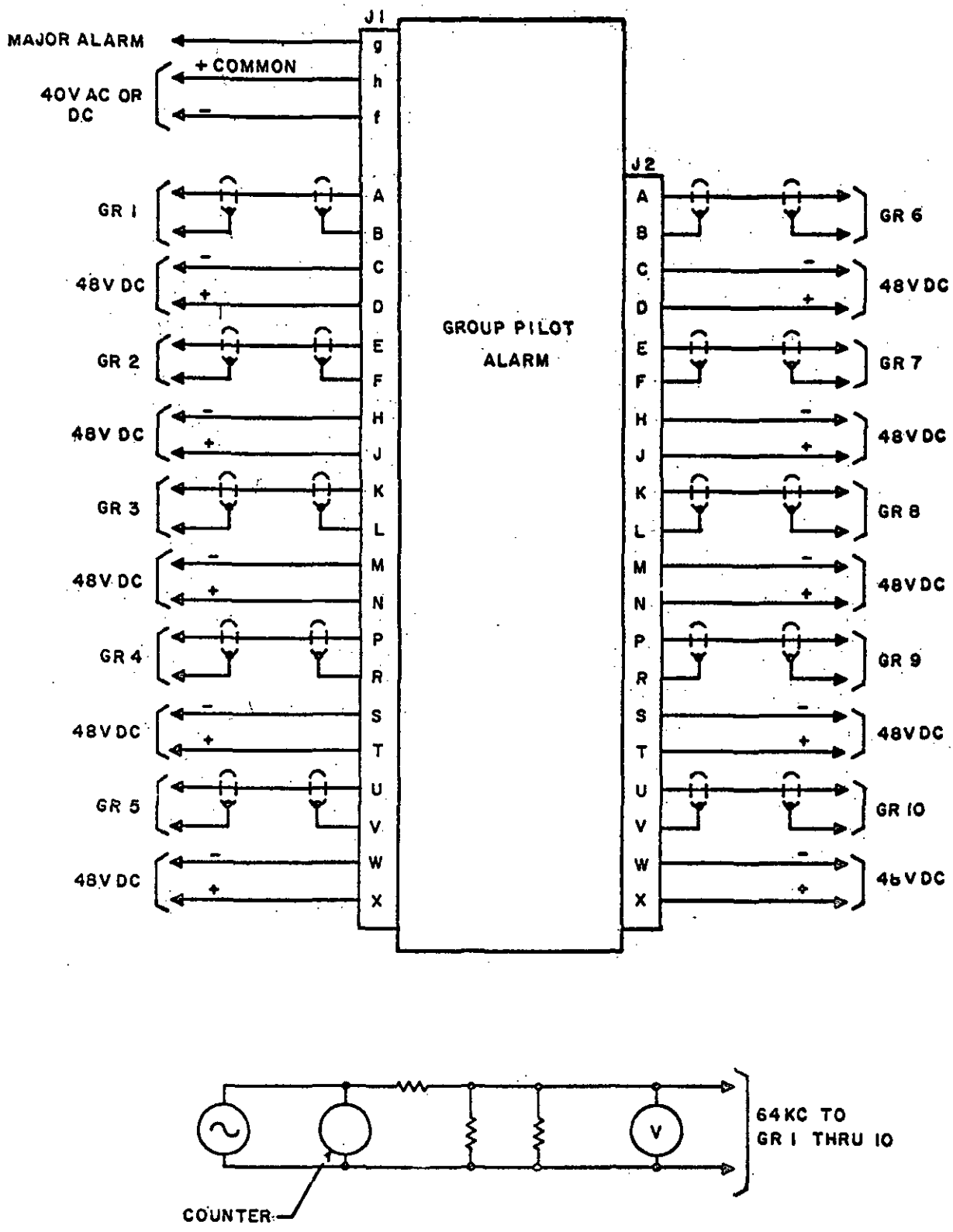


Figure 1. Test Setup (Typical)

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TABLE I

TEST NO	TEST CONDITIONS			TEST METHOD	REQUIREMENTS			
	ALL ALARM CUTOFF SWITCHES	48 v dc	PILOT INPUT 64 kc		GROUP PILOT ALARM LAMP	ALARM CUTOFF LAMP	MAJOR ALARM (OHMMETER)	48 v dc SUPPLY CURRENT (ma)
1	NORMAL	OFF	OFF		ON	OFF	SHORT	- - -
2	NORMAL	OFF	OFF	Switch each ALARM CUTOFF switch in turn to CUTOFF and then back to NORMAL.	ON	ON and then back to OFF for each group	SHORT	- - -
3	CUTOFF	OFF	OFF	Switch each ALARM CUTOFF switch in turn to NORMAL and then back to CUTOFF.	ON	ON	SHORT and then back to OPEN	- - -
4	CUTOFF	ON*	ON*	Switch ALARM CUTOFF switch to NORMAL (only on group being tested).	OFF (only for group being tested)	ON	OPEN	18 ± 1
5	REPEAT TEST 4 FOR EACH GROUP IN SUCCESSION							

* Apply to one group at a time

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4.6.5 RF interference test.- The group pilot alarm shall be tested in accordance with Class G_p of MIL-STD-826. When out of specification readings are noted, formal testing will stop, fixes applied and retest made until equipment meets the requirements.

4.6.6 Maintainability demonstration.- The maintainability of the group pilot alarm shall be demonstrated in accordance with Appendix A of MIL-M-26512 to a confidence level of 90 percent.

4.6.7 The reliability of the equipment shall be demonstrated in accordance with the test method presented in MIL-R-26474. The test shall be performed for sufficient time to permit a decision to accept or reject the equipment in accordance with Table 1 of MIL-R-26474.

4.6.7.1 Production reliability testing.- A production reliability test shall be performed on a quantity of production equipments in accordance with MIL-R-26474.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instruction from the procuring activity.

6. NOTES

6.1 Intended use. The group pilot alarm shall detect a 64-KC pilot signal and supply a major alarm indication to Control Indicator C-3688( )/MCC-12, Indicator Power Supply ID-1077( )/MRC-98.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable level of preservation, packaging and packing required. (See 5.1)

c. Whether the testing will be conducted by the procuring activity or by the contractor. (See 4.1)

d. Conditions under which design approval will be granted. (See 3.4)

e. Whether the finish and color will be other than as specified herein. (See 3.9)

f. Whether a preprpduction test report will be submitted by the contractor.(See 4.4.2)

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g. Whether nomenclature and nameplate data will be required of the contractor and when this data will be provided to the procuring activity. (See 3.3.6)

h. Whether a reliability analysis will be submitted by the contractor (see 3.3.8 c) and time of submission.

i. Whether a revised reliability analysis will be submitted by the contractor (see 3.3.8 c(1)) and when it shall be submitted.

j. Whether a reliability report will be submitted by the contractor (see 3.3.8 c(2)) and when it shall be submitted.

k. Whether a final reliability will be submitted by the contractor (see 3.3.8 c(3)) and when it shall be submitted.

l. The quantity of preproduction samples to be tested. (See 4.4.1)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*: RADC (EMLES)

CITY AND STATE: GRIFFISS AFB, NEW YORK

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT: $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. *COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID*

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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